

# Wages and Employment in a Single Labour Market

ECON 300 - Labour Economics

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# Learning Objectives

- Construct market supply and demand and compute equilibrium wage and employment.
- Contrast competitive outcomes with imperfect competition (monopoly, oligopoly, monopolistic competition).
- Explain how monopsony reduces wages and employment relative to competition.

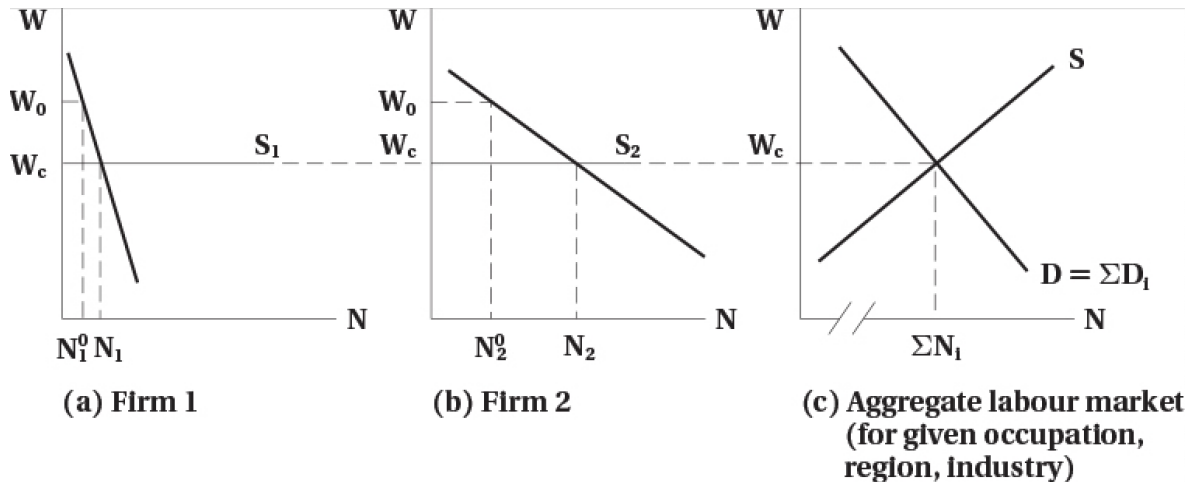
# Main Questions

- How are wages and employment determined in a single labour market?
- How does imperfect competition modify the supply-demand analysis?
- Who bears payroll taxes – workers or firms?
- What is monopsony? How different are outcomes when firms have hiring power?

# The Competitive Firm's Interaction with the Market

- Competitive in both product and labour markets
- Homogeneous labour; firm is a price and wage taker
- Firm can hire all needed labour at the market wage
- Market wage set by aggregate labour market

Figure 7.1: Competitive Product and Labour Markets



# Panels (a) and (b): Individual Firm Labour Demand

## Firm Labour Demand

- Downward-sloping curve = **value of marginal product of labour** ( $VMP_N$ )
- Hiring rule:

$$\text{Hire labour until } VMP_N = W$$

- Each firm faces a **perfectly elastic labour supply** at the market wage

## Effect of a Wage Change

- At wage  $W_0$ , Firm  $i$  hires  $N_i^0$
- At lower wage  $W_c$ , labour hired increases to  $N_i$
- Firms hire more labour when wages fall

## Key Insight

- Individual firms take wages as given
- Labour demand responds to wage changes

## Panel (c): Aggregate Labour Market

### Market Labour Demand

- Market demand is the **horizontal sum** of firm demands:

$$D = \sum_i D_i$$

- At wage  $W_0$ :

$$N^{market} = N_1^0 + N_2^0 + \dots$$

### Labour Supply

- Upward-sloping curve  $S$  represents market labour supply

### Equilibrium

- Wage  $W_c$  clears the market
- Labour supply equals labour demand
- Firms hire where  $W_c = VMP_N$

# Labour Supply in Figure 7.1

## Labour Supply at the Firm Level (Panels a & b)

- Each firm faces a **perfectly elastic labour supply** at the market wage
- Represented by horizontal lines ( $S_1$ ,  $S_2$ )
- Reason:
  - Individual firms are wage takers
  - They can hire as many workers as desired at the prevailing wage
  - Hiring decisions do *not* affect the wage

## Labour Supply in the Market (Panel c)

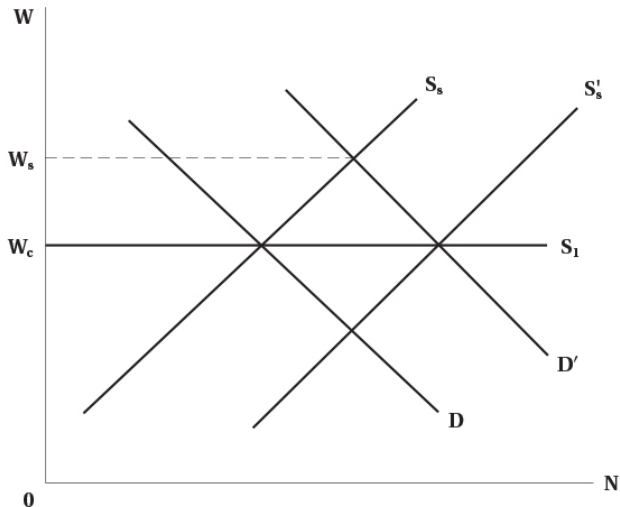
- Market labour supply curve ( $S$ ) is **upward sloping**
- Reflects workers' willingness to supply more labour at higher wages
- Determined by:
  - worker preferences (income vs leisure)
  - alternative employment opportunities
  - demographics and participation decisions



# Short Run vs. Long Run to the Firm

- Short run: Product demand rise  $\Rightarrow$  higher labour demand; wage and employment rise (given upward-sloping supply to the firm).
- Long run: Higher short-run wages attract more workers; labour supply shifts right; wage adjusts.

Figure 7.2: Short-Run and Long-Run Labour Supply Schedules



## Short-run:

- Even a competitive firm may have to raise its wages to attract additional workers
  - dynamic monopsony
- The firm's short-run labour supply schedule could be upward sloping,  $S_s$

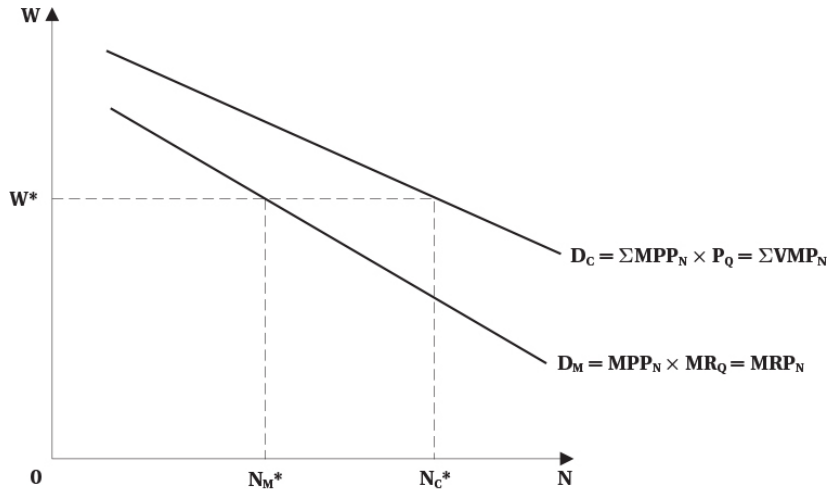
## Long-run:

- Since  $W_s > W_c \Rightarrow$  there will be supply influx of additional workers
- Resulting in the shift of supply schedule from  $S_s$  to  $S'_s$
- The supply influx would depress the temporarily high, short-run wage of  $W_s$  back to its long-run level of  $W_c$

# Monopoly: Effects of Hiring More Labour

- $MPP_N \downarrow$  with more  $N$
- $MR$  falls as more output sold requires lower price
- Labour demand under monopoly  $<$  under competition

Figure 7.3: Monopolist vs. Competitive Demand for Labour



## Figure 7.3: Monopolist vs Competitive Demand for Labour

### Competitive Labour Demand ( $D_C$ )

- Firms are price takers in the product market
- Output price is constant:  $P_Q$
- Labour demand equals the **value of marginal product**:

$$D_C = VMP_N = MPP_N \times P_Q$$

- Higher employment level:  $N_C^*$

### Monopolist Labour Demand ( $D_M$ )

- Marginal revenue is less than price:  $MR_Q < P_Q$
- Labour demand equals the **marginal revenue product**:

$$D_M = MRP_N = MPP_N \times MR_Q$$

- Curve lies below  $D_C$  and is steeper –  $MR_Q$  falls as employment and output increase
- Lower employment level:  $N_M^*$

# Oligopoly & Monopolistic Competition

- Oligopoly: few firms; strategic interdependence
- Possible departures from market wages (above-normal profits may be partly shared)
- Monopolistic competition: short-run profits  $\Rightarrow$  possibly higher-than-market wages; long-run zero profit  $\Rightarrow$  market wages

# Linear Supply and Demand: Setup

$$N_D = a + bW; b < 0, \quad N_S = c + fW; f > 0$$

Setting  $N_S = N_D$  yields equilibrium  $(w^*, N^*)$ .

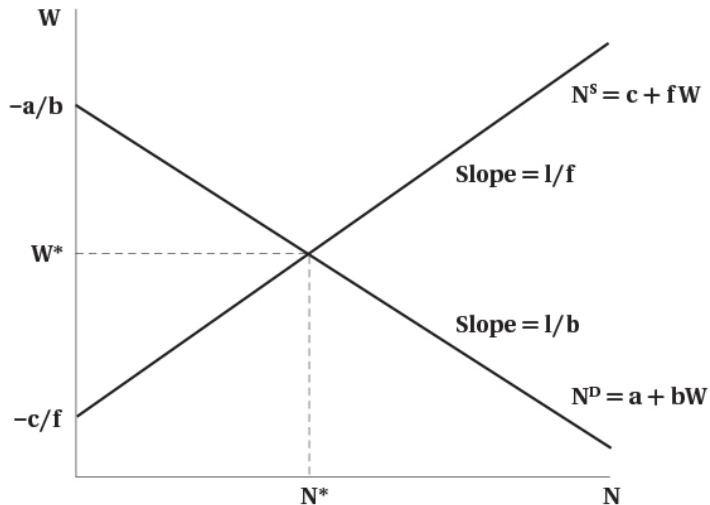
**With Shifter  $z$ :**

$$N_D = a + bw + \gamma z, \quad N_S = c - dw + \delta z$$

Solve again for new  $(w^*, N^*)$  to study incidence and comparative statics.



Figure 7.4: Linear Supply and Demand



## Application: Incidence of a Unit Payroll Tax

- Examples: CPP/QPP, workers' compensation, EI, health/education payroll levies
- Often called “job killers” — but actual incidence depends on elasticities

TABLE 7.1: Payroll Taxes as a Percentage of Labour Income, Canada (1971, 1997)

| Type of Payroll Tax     | 1971        | 1997         |
|-------------------------|-------------|--------------|
| Workers' compensation   | 0.73        | 1.61         |
| Unemployment insurance  | 1.06        | 4.98         |
| Canada/Quebec pension   | 2.12        | 3.92         |
| Health and/or education | 0.21        | 1.71         |
| <b>Total</b>            | <b>4.12</b> | <b>12.23</b> |

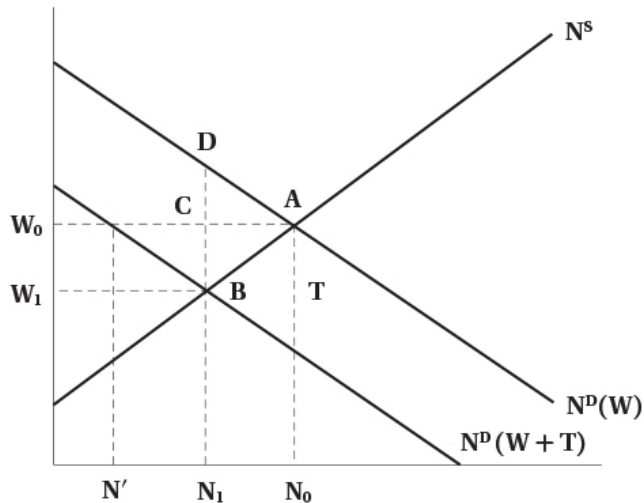
# Payroll Tax in the Model

Let a per-worker tax  $t$  on firms shift the perceived cost:

$$N_D = c - d(w + t)$$

Incidence on wages vs. employment depends on the elasticities of  $N_S$  and  $N_D$ .

Figure 7.5: Effect of a Payroll Tax on Employment and Wages



## Figure 7.5: Payroll Tax, Wages, and Employment

### Initial Equilibrium

- Labour supply:  $N^S$
- Labour demand:  $N^D(W)$
- Initial equilibrium at point  $A$
- Wage =  $W_0$ , Employment =  $N_0$

### Introduction of a Payroll Tax ( $T$ per worker)

- Tax raises cost of labour to employers to  $W + T$
- Labour demand shifts downward to  $N^D(W + T)$
- New equilibrium at point  $B$
- Workers receive lower wage:  $W_1$
- Employment falls from  $N_0$  to  $N_1$

## Figure 7.5: Payroll Tax, Wages, and Employment

### Tax Incidence

- Total tax per worker =  $T$
- Workers bear part of the burden:

$$\text{Worker share} = BC$$

- Employers bear the remainder:

$$\text{Employer share} = CD$$

### Summary

- Payroll taxes reduce employment and wages
- Tax burden is shared between workers and firms
- The side of the market that is less elastic bears more of the tax burden

# Monopsony: Basics

- A monopsony is a labour market with one dominant firm.
- Large firm relative to labour market size; influences wage
- It faces an upward-sloping labour supply curve
- To hire more workers, the firm must offer higher wages.
- Marginal cost of labour exceeds average cost (wage) due to **inframarginal uprating**

**Inframarginal uprating** is an increase in benefit levels for individuals already eligible for a program.

- Eligibility rules remain unchanged.
- Raises payments to current recipients
- Does not expand eligibility
- Does not directly add new participants



## Example

- Unemployment benefits increase from \$500 to \$550 per week for current recipients.

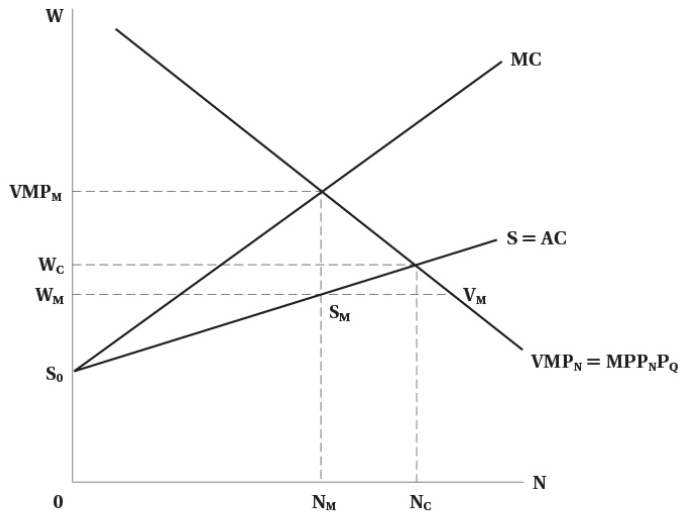
## Economic Effects

- Increases recipients' income
- May reduce labour supply through an income effect
- Participation incentives at eligibility thresholds remain unchanged

## Contrast

- **Marginal reform:** A policy that expands eligibility or lowers the cutoff – affects marginal entrants, not inframarginal recipients.

Figure 7.6: Monopsony



## Figure 7.6: Monopsony in the Labour Market

### Curves in the Figure

- Labour supply ( $S = AC$ ): wage required to attract workers; also the average cost of labour.
- Marginal cost of labour ( $MC$ ): lies above supply since hiring one more worker requires raising wages for all workers.
- Labour demand ( $VMP_N$ ):

$$VMP_N = MPP_N \times P_Q$$

downward sloping due to diminishing productivity.

### Monopsony Outcome

- Profit maximization occurs where  $MC = VMP_N$ .
- Employment:  $N_M$
- Wage paid:  $W_M$  (from the supply curve)

## Figure 7.6: Monopsony in the Labour Market

### Competitive Benchmark

- Competitive equilibrium:  $N_C$  and  $W_C$
- Monopsony results:

$$N_M < N_C \quad \text{and} \quad W_M < W_C$$

### Key Point

- Market power allows the firm to restrict employment and suppress wages.

# Perfect Monopsonistic Wage Differentiation

## Definition

- Occurs when a monopsonist pays each worker a wage equal to their **reservation wage**.
- The firm has full information about workers' minimum willingness to work.
- Workers receive different wages instead of a single uniform wage.

## Key Mechanism

- Hiring an additional worker does *not* require raising wages for existing workers.
- The labour supply curve becomes the firm's marginal cost of labour:

$$MC = \text{Labour Supply}$$

## Hiring Rule

- The firm hires labour where:

$$VMP_N = \text{Labour Supply}$$

## Employment Outcome

- Employment equals the competitive level.
- The efficient number of workers is hired.

## Distribution of Surplus

- Workers receive wages equal to their reservation wages.
- Worker economic rent (the surplus income a worker receives above their reservation wage) is eliminated.
- The firm captures all surplus.

## Comparison

- Competitive market: single wage, surplus shared.
- Single-wage monopsony: lower wage, reduced employment.
- Perfect wage discrimination: efficient employment, firm captures surplus.

## Real-World Relevance

- Pure wage discrimination is rare.
- Partial forms appear through:
  - individualized contracts
  - performance pay
  - bargaining and bonuses
  - non-wage compensation

- Competitive equilibrium determines wage and employment; product market power lowers labour demand.
- Payroll tax incidence depends on elasticities; wages and employment adjust accordingly.
- Monopsony lowers wage and employment; targeted minimum wage can raise both.



# Thank You!